

# RAJASEKHAR REDDY KOTHAPALLI

Software Engineer | Java | Backend & Distributed Systems

Open to Relocation • +1 (913)-413-4174 • [rxk40660@ucmo.edu](mailto:rxk40660@ucmo.edu) • [GitHub](#) • [LinkedIn](#) • [Portfolio](#)

---

## Summary

Software Engineer with 2 years of professional experience at Zoho developing production Java backend and distributed systems for large-scale VOD, livestreaming, and WebRTC platforms. Experienced in Java, Spring Boot, REST APIs, microservices, distributed systems, cloud technologies, CI/CD, and performance optimization. Improved video fetching and rendering performance by 40% through backend file-processing optimizations. M.S. in Computer Science with hands-on experience building scalable backend applications and performance-testing distributed services.

## Work experience

---

### Member Technical Staff, Zoho

Aug 2022 - Aug 2024

- Java backend development for VOD, Livestreaming & WebRTC platforms
  - Developed and optimized **Java backend services** for **Video-on-Demand (VOD), livestreaming, and WebRTC platforms**, supporting scalable media delivery and distributed file-processing workflows.
  - Engineered advanced **Java file-fetching and JSON processing modules**, reducing video fetching and rendering time by **40%** and improving overall streaming performance.
  - Implemented **Java-based GZIP compression and data optimization mechanisms** for VOD services, reducing network transmission overhead and improving content-delivery efficiency.
  - Developed media-delivery services using **Fragmented MP4 (fMP4)** and optimized **segment management and adaptive bitrate streaming (ABR)** workflows to reduce startup latency and improve playback performance across varying network conditions.
  - Designed and implemented distributed **Point of Presence (PoP) services in Java**, improving content availability, scalability while reducing delivery latency across geographically distributed environments.
  - Built **Python/Locust load-testing suites** to simulate concurrent workloads, measure backend performance, identify bottlenecks, and validate scalability of distributed services.
  - Developed Java-based **MetaFile Manager and FileWatcher components** for distributed file systems, enabling real-time file discovery, monitoring, and efficient retrieval across networked environments.
  - Implemented **checksum-based file-integrity validation** to detect data corruption or tampering and improve reliability of file transfers across distributed systems.
  - Enhanced **File-Copy and File-Sync** services within WebRTC workflows, improving file-transfer performance, reliability, and synchronization accuracy across distributed systems.

---

## Projects

- **WhisperFlowApp – Real-Time Speech-to-Text:** Developed a privacy-focused, **offline speech-to-text desktop application** using **C#, .NET 8, WPF, NAudio, and Whisper.cpp**, supporting real-time audio capture and local transcription without sending user data to external services.
- **Secure Chat Application:** Built a **Java-based real-time messaging application** using **WebSockets, JSON, and AES-256 encryption**, enabling encrypted, anonymous, registration-free client-server communication.

## Projects

---

- **Appliance Warehouse:** Developed a **full-stack e-commerce web application** using **HTML, CSS, JavaScript, and Flask**, supporting online browsing, purchasing, and weekly/monthly rental workflows for household appliances.
- **Custom Python UNIX Shell:** Designed and implemented a **UNIX-like command-line shell in Python** with command execution, **piping, I/O redirection, and command auto-completion**, demonstrating operating-system and process-management fundamentals.
- **Battery Alarm:** Developed an **Android application in Java** that monitors battery status and alerts users when the device reaches a configured/full charge level.
- **TrustVault – Decentralized Digital Ownership Platform:** Developed a decentralized proof-of-ownership concept using **IPFS and the Sui blockchain**, enabling verifiable ownership records for digital content through distributed storage and blockchain-based verification.

## Technical Skills

---

- **Programming Languages :** Java (8/11/17/21), Python, C#, C, JavaScript, SQL
- **Backend & API Development:** Spring Boot, Spring MVC, Spring Cloud, Spring Security, Spring Data JPA, Hibernate, REST APIs, RESTful Web Services, Microservices, WebSockets, gRPC, OAuth 2.0, JWT, API Integration, Client-Server Architecture
- **Web & Application Development:** HTML5, CSS3, JavaScript, React.js, Node.js, Bootstrap, Flask, Django, .NET, WPF
- **Cloud & DevOps:** AWS, Docker, GitHub Actions, Azure DevOps, CI/CD
- **Databases & Storage:** MySQL, PostgreSQL, Oracle, MongoDB, Redis
- **Testing & Performance:** Unit Testing, Integration Testing, Load Testing, Performance Testing, Locust, Performance Optimization
- **Version Control & Developer Tools:** Git, GitHub, Maven, Jira, Postman, IntelliJ IDEA, Visual Studio
- **Software Engineering:** Object-Oriented Programming (OOP), Data Structures & Algorithms, Design Patterns, SDLC, System Design, Distributed Systems, Multithreading, Debugging, Code Reviews, Agile/Scrum

## Education

---

### Master of Science in Computer Science | University of Central Missouri

- Conducted blockchain-focused research under faculty supervision as a Research Assistant.

### Bachelor of Technology in Computer Science | Krishna University

## Certifications & Professional Development

---

Aug 2024 - May 2026

- **Python for Data Science – Elite + Silver Certification** | NPTEL (National Programme on Technology Enhanced Learning). Aug 2019 - May 2022
- **Diploma in Hardware & Networking – A+ Grade** | International Council of Education & Training (IJET).